```
# Install required libraries (Run only in Google Colab)
# !pip install scikit-learn pandas numpy matplotlib

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# ------------------------------
# Step 1: Create Sample Dataset
# ------------------------------

np.random.seed(42)
size = 200

data = pd.DataFrame({
    "Income": np.random.randint(20000, 100000, size),
    "LoanAmount": np.random.randint(50000, 300000, size),
    "CreditScore": np.random.randint(300, 850, size),
    "EmploymentYears": np.random.randint(0, 15, size)
})

# Loan Approval Rule (Target Variable)
data["Loan_Status"] = (
    (data["Income"] > 40000) &
    (data["CreditScore"] > 600) &
    (data["EmploymentYears"] > 2)
).astype(int)
```

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# ------------------------------
# Step 2: Split Data
# ------------------------------

X = data.drop("Loan_Status", axis=1)
y = data["Loan_Status"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ------------------------------
# Step 3: Train Model
# ------------------------------

model = RandomForestClassifier()
model.fit(X_train, y_train)

# ------------------------------
# Step 4: Accuracy
# ------------------------------

y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", round(accuracy * 100, 2), "%")

# ------------------------------
# Step 5: Take User Input
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print("\nEnter Applicant Details:")

income = int(input("Enter Annual Income: "))
loan_amount = int(input("Enter Loan Amount: "))
credit_score = int(input("Enter Credit Score (300-900): "))
employment_years = int(input("Enter Employment Years: "))

new_data = pd.DataFrame({
    "Income": [income],
    "LoanAmount": [loan_amount],
    "CreditScore": [credit_score],
    "EmploymentYears": [employment_years]
})

prediction = model.predict(new_data)[0]

if prediction == 1:
    print("\nLoan Approved ✅")
else:
    print("\nLoan Rejected ❌")

# ------------------------------
# Step 6: Explainable AI (Feature Importance)
# ------------------------------

importances = model.feature_importances_
features = X.columns

plt.bar(features, importances)
plt.title("Feature Importance (Explainable AI)")
plt.xlabel("Features")
```

```
plt.ylabel("Importance")
plt.show()
```